

MKT3434 MACHINE LEARNING

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Date: March 27, 2025

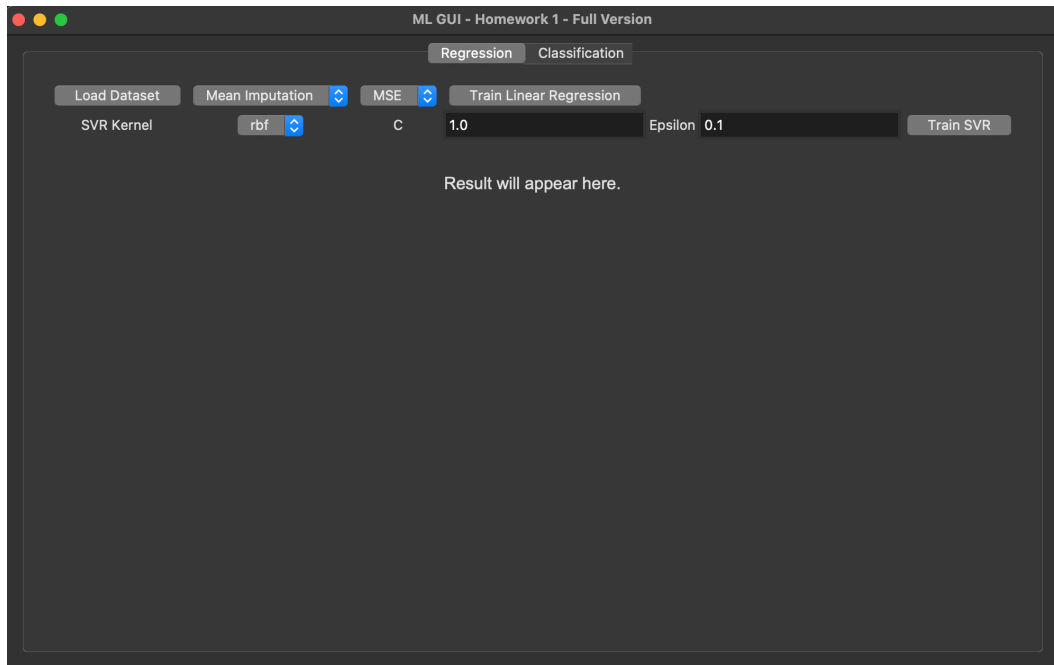
Project: #1: GUI Enhancements for Data Handling and Model Completion

1. Summary

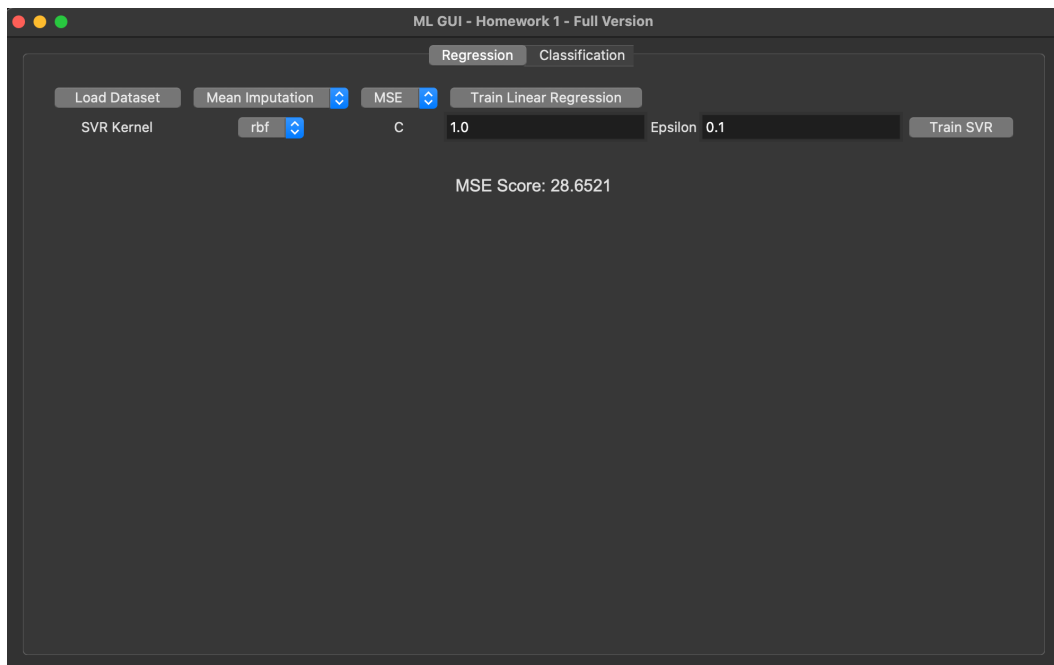
This project is a GUI-based machine learning model trainer, developed as part of Homework Assignment #1. The GUI supports both regression and classification tasks, allows users to load datasets, handle missing data through various methods, select machine learning models (Linear Regression, SVR, Logistic Regression, SVM), and choose loss functions appropriate for each task. The results are displayed directly on the interface after training.

The regression tasks were tested using the Boston Housing dataset, and classification tasks were tested using the Iris dataset.

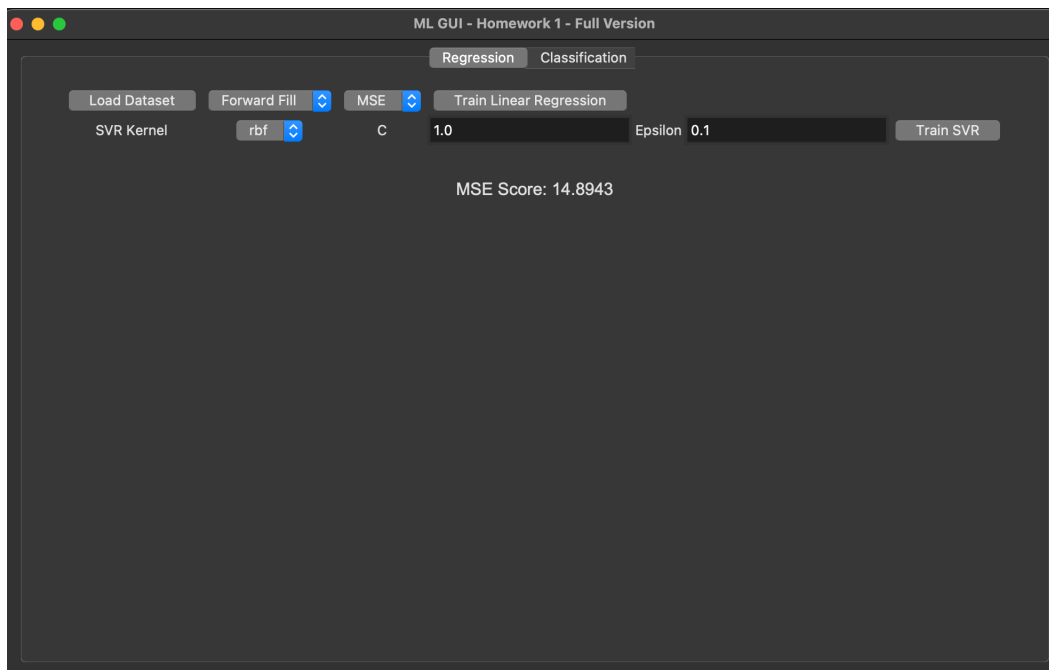
2. Screenshots & Results



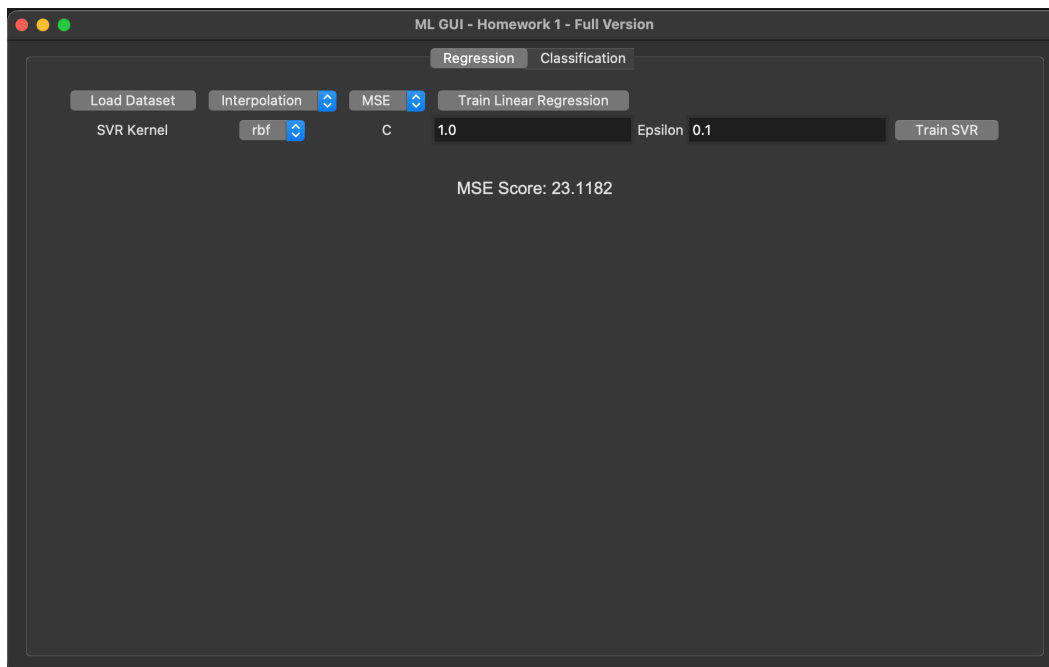
Initial GUI layout before loading any dataset.



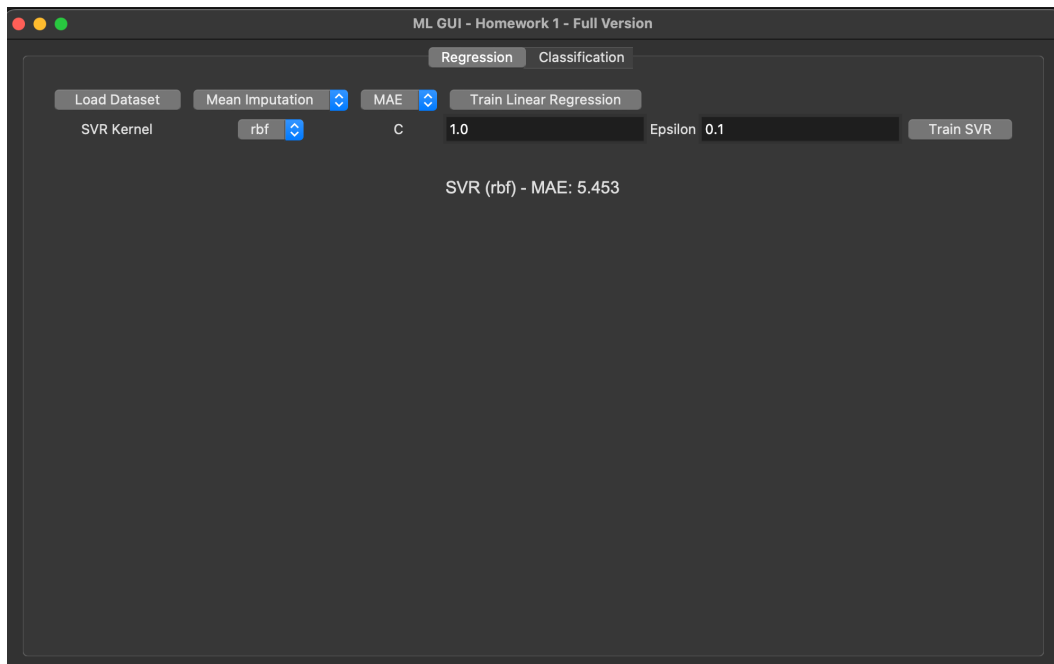
Linear Regression trained using Mean Imputation and MSE Loss.



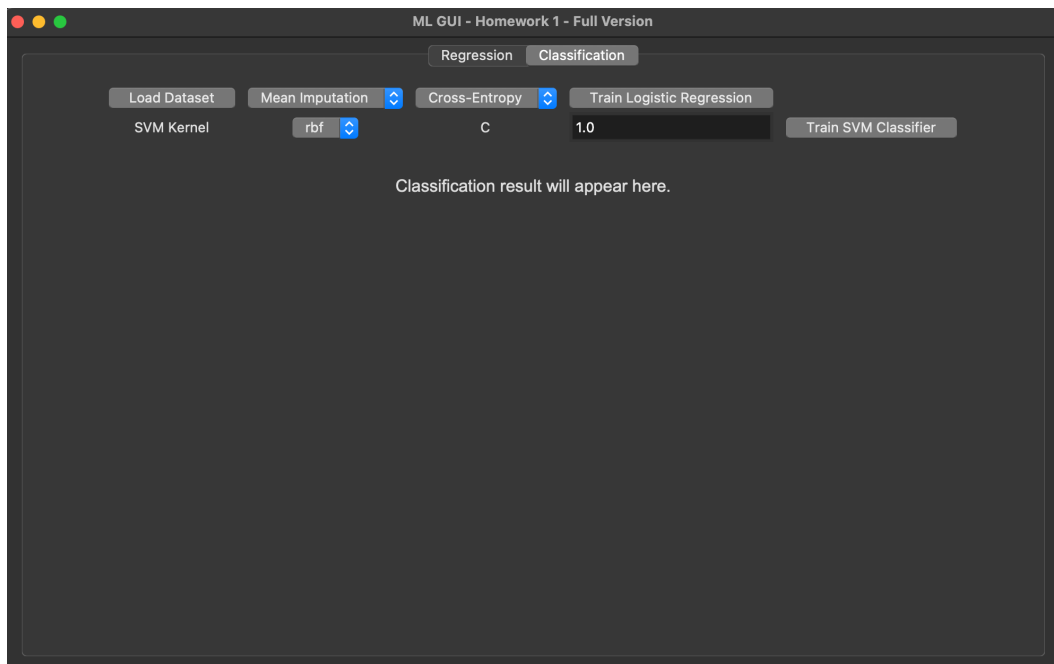
Linear Regression with Forward Fill shows improved MSE.



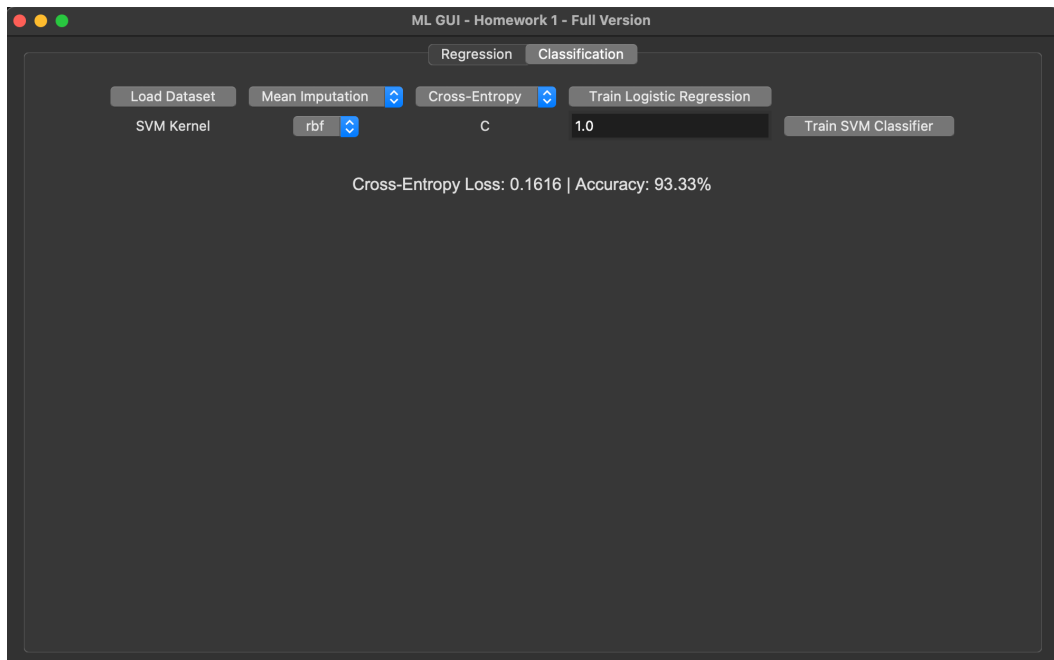
Linear Regression trained using Interpolation method.



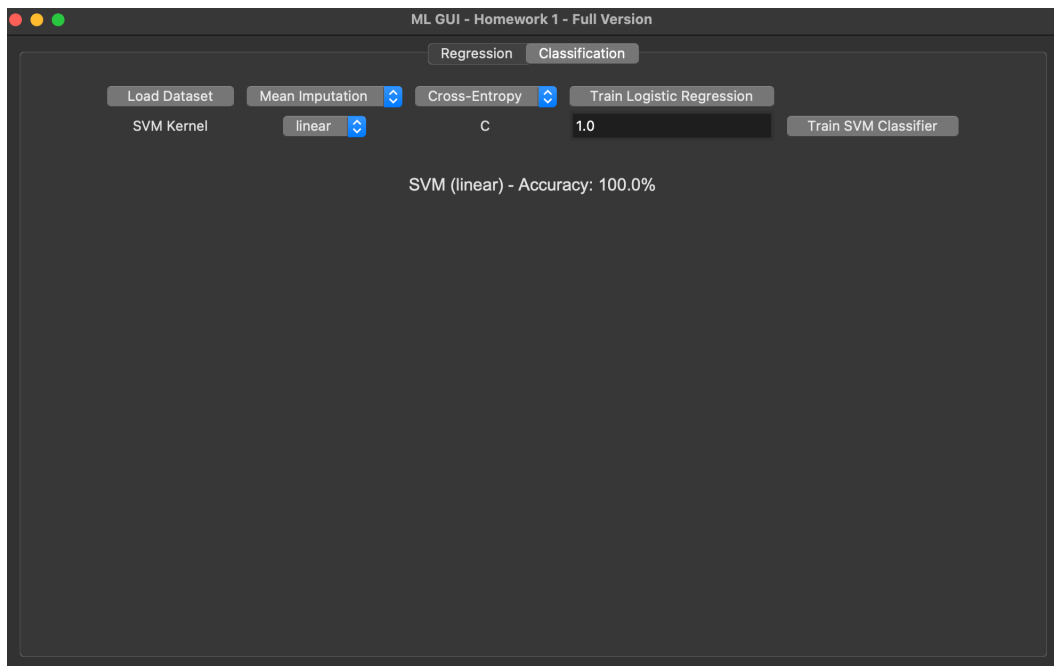
SVR trained with RBF kernel and MAE loss function.



Classification tab interface before training.



Logistic Regression model trained using Cross-Entropy loss.



SVM Classifier trained with linear kernel and achieved 100% accuracy.

3. Comparison of Missing Data Handling

The following MSE values were observed for Linear Regression using the Boston Housing dataset with different missing data handling methods:

- Mean Imputation → MSE: 28.65
- Forward Fill → MSE: 14.89
- Interpolation → MSE: 23.12

As shown, Forward Fill provided the best regression performance in this scenario. This highlights the impact of preprocessing methods on model accuracy and effectiveness.